

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2023-24

Paper No. BPT-301: Optics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **five** of the following: [5 × 4 = 20]

- (a) In a Young's double-slit experiment, interference fringes are formed using sodium light comprising two wavelengths 5890 Å and 5896 Å. Obtain the regions on the screen where the fringe pattern will disappear. Assume $d = 0.5$ mm and $D = 100$ cm.
- (b) A plane-polarized light is incident perpendicularly on a quartz plate cut with faces parallel to the optic axis. Find the thickness of the quartz plate which introduces a phase difference of 60° between the e and o rays.
- (c) A left circularly polarized beam ($\lambda_0 = 5893$ Å) is incident normally on a calcite crystal (with its optic axis cut parallel to the surface) of thickness 0.005141 mm. What will be the state of polarization of the emergent beam? For calcite, $n_o = 1.65836$ and $n_e = 1.48641$.
- (d) In a double-slit diffraction experiment, the slit width $b = 7 \times 10^{-3}$ cm, the separation between identical points of the two slits $d = 7 \times 10^{-2}$ cm, and $\lambda = 6300$ Å. How many interference minima will occur within the first diffraction minima on either side of the central diffraction maximum? Which orders of interference maxima will be missing?
- (e) Show that a zone plate acts like a converging lens with multiple foci.
- (f) How can the central fringe be seen in a Lloyd's mirror experiment? Show that the central fringe is sharper in a Lloyd's mirror than in a Fresnel biprism experiment.
- (g) White light is reflected normally from a uniform oil film ($n = 1.33$). An interference maximum for 6000 Å and a minimum for 4500 Å coincide. Calculate the thickness of the film.

Question 2.

- (a) In a Michelson interferometer experiment performed with a source which consists of two wavelengths 4882 Å and 4886 Å, through what distance does the mirror have to be moved between two successive positions of the disappearance of the fringes? [6]
- (b) Discuss the intensity distribution of a Fabry-Pérot interferometer and show that the larger the value of R (reflectivity of the mirrors), the greater the difference between I_{max} and I_{min} . In what way is it superior to the Michelson interferometer? [8]

Question 3.

- (a) Deduce the expression of the intensity distribution of Fraunhofer diffraction due to double slits. Discuss the nature of the intensity distribution and explain how the intensity falls off to zero on either side of the central fringe and gradually reappears. [8]
- (b) In a Newton's ring arrangement with $\lambda = 6.4 \times 10^{-5}$ cm and the radius of curvature of the convex surface is 100 cm, if the lens is vertically raised up, discuss the change in the ring pattern as seen through the microscope. [6]

OR

- (a) Consider a diffraction grating of width 5 cm with slits of width 0.0001 cm separated by a distance of 0.0002 cm. How many orders would be observable for $\lambda = 5.5 \times 10^{-5}$ cm? Calculate the width of the principal maximum. Would there be any missing orders? For this diffraction grating, calculate the dispersion and resolving power in the different orders. [8]
- (b) What is the Rayleigh criterion of resolving power? Find the resolving power of a grating. Discuss what happens if the number of grating elements is too large. [6]

Question 4.

- (a) You are given a Nicol prism and a quarter-wave plate. Explain the method to produce and detect plane, elliptically, and circularly polarized light. [10]
- (b) What are the states of polarization when the x - and y -components of the electric field are given by the following equations:

(i) $E_x = E_0 \cos(\omega t + kz), \quad E_y = \frac{E_0}{2} \cos(\omega t + kz + \pi)$

(ii) $E_x = E_0 \sin(\omega t + kz), \quad E_y = E_0 \cos(\omega t + kz)$

[4]

OR

- (a) Explain birefringence. What are quarter- and half-wave plates? What are their uses? [6]
- (b) Discuss the working of a biquartz polarimeter. [8]

Question 5.

- (a) What are the differences between spontaneous and stimulated emission? Derive the relations between the Einstein coefficients. [8]
- (b) The length of a laser tube is 150 mm and the gain factor of the laser material is 0.0005 per cm. If one of the cavity mirrors reflects 100 percent of the light, what is the required reflectance of the other cavity mirror? [6]

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